

Finitely Many Points on the Line Under the Influence of an Exponential Potential — An Integrable System

Author: Jürgen Moser • Courant Institute of Mathematical Sciences, New York University, New York, NY 10012

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ABSTRACT

We study the classical dynamics of n particles on a one-dimensional real line interacting through exponential repulsive nearest-neighbor potentials (the non-periodic Toda lattice). Using the transformation introduced by Hermann Flaschka, the equations of motion are cast into a Lax pair differential equation $dL/dt = [B, L]$ on symmetric tridiagonal Jacobi matrices. We prove that the system is completely integrable in the sense of Liouville. In the asymptotic limit $t \rightarrow +\infty$, the particle interactions decay exponentially ($a_k \rightarrow 0$), and the diagonal elements $b_k(t)$ decouple and converge to the system's conserved eigenvalues arranged in strictly descending order. Thus, the physical scattering of particles in an exponential potential naturally executes a continuous sorting algorithm.

1. Hamiltonian of the Non-Periodic Toda Lattice

Consider n unit-mass particles on the real line with coordinates $q_1 < q_2 < \dots < q_n$ and momenta $p_1, \dots, p_n$.

The Hamiltonian is given by:

$$H(p, q) = \frac{1}{2} \sum_{k=1}^n p_k^2 + \sum_{k=1}^{n-1} \exp(-(q_{k+1} - q_k))$$

The Hamilton equations of motion are:

$$dq_k/dt = dH/dp_k = p_k$$

$$dp_k/dt = -dH/dq_k = \exp(-(q_k - q_{k-1})) - \exp(-(q_{k+1} - q_k))$$

with boundary conventions $q_0 = -\infty$ and $q_{n+1} = +\infty$.

2. The Flaschka-Moser Coordinate Transformation

In 1974, Hermann Flaschka introduced the change of coordinates:

$$a_k = \frac{1}{2} \exp(-(q_{k+1} - q_k)/2), \quad k = 1, \dots, n-1$$

$$b_k = -\frac{1}{2} p_k, \quad k = 1, \dots, n$$

In these coordinates, the phase space equations become polynomial:

$$da_k/dt = a_k (b_{k+1} - b_k), \quad k = 1, \dots, n-1$$

$$db_k/dt = 2 (a_k^2 - a_{k-1}^2), \quad k = 1, \dots, n$$

with $a_0 = a_n = 0$.

3. The Tridiagonal Lax Pair

Define the real symmetric tridiagonal Jacobi matrix L and skew-symmetric matrix B :

$$L = \text{tridiag}(a_k, b_k, a_k), \quad B = \text{tridiag}(a_k, 0, -a_k)$$

Then the Toda equations of motion are identically equivalent to the Lax equation:

$$dL/dt = [B, L] = B L - L B$$

Theorem (Liouville Integrability): The eigenvalues $\lambda_1, \dots, \lambda_n$ of L are rigorous first integrals of motion (conserved quantities) in involution: $\{Tr(L^k), Tr(L^m)\} = 0$.

4. Asymptotic Scattering and Eigenvalue Sorting

Theorem (Moser Asymptotic Scattering): As $t \rightarrow +\infty$, all particles separate indefinitely: $q_{k+1} - q_k \rightarrow +\infty$.

Consequently, the off-diagonal terms $a_k(t)$ decay exponentially to zero as $t \rightarrow +\infty$:

$$\lim_{t \rightarrow +\infty} a_k(t) = 0$$

Therefore, the Jacobi matrix $L(t)$ becomes strictly diagonal:

$$\lim_{t \rightarrow +\infty} L(t) = \text{diag}(b_1(\infty), b_2(\infty), \dots, b_n(\infty)) = \text{diag}(\lambda_1, \dots, \lambda_n)$$

Because faster particles must move ahead of slower particles in 1D space, the asymptotic velocities are strictly sorted: $\lambda_1 > \lambda_2 > \dots > \lambda_n$. Reversing time ($t \rightarrow -\infty$) produces ascending order: $\lambda_1 < \lambda_2 < \dots < \lambda_n$!

The non-periodic Toda lattice is therefore a completely integrable physical sorting machine.